Pulkit Garg

pulkit3110@gmail.com | +91-9315307577 | LinkedIn | GitHub | pulkitgarg31.tech

Area of Expertise

Backend Development · Scalable Systems · REST APIs · Data Structures & Algorithms · System Design · Agentic AI

Education

Thapar Institute of Engineering and Technology, Patiala Aug 2023 – May 2027
Bachelor of Engineering in Computer Science and Engineering *CGPA: 8.71*

Projects

SynthSales – Agentic AI Sales CRM Live |
Tech: FastAPI, Gemini API, PostgreSQL, Redis, Docker, Google APIs, APScheduler

Agentic B2B outreach backend that turns uploaded company CSVs into researched, scored, contactable prospects with verified emails, AI-drafted outreach, reply tracking, and meeting scheduling.

- Built a scalable FastAPI backend with 78 REST endpoints and 14 PostgreSQL models, owning it end-to-end from design to JWT auth and admin approvals.
- Designed an orchestrator that runs 8 AI agents in order, keeps a 24h undo snapshot before any run that clears data, and protects live conversations, booked meetings and verified emails from re-runs.
- Added an opt-in autonomous mode where AI reads inbound replies, answers product questions and books meetings, acting only above a confidence floor and sending the rest to human review.

Real-Time Self-Healing Data Pipeline
Tech: Python, Apache Airflow, Ollama LLaMA 3.2, pytest, PostgreSQL, dbt, Docker, Streamlit

Fault-tolerant Airflow pipeline that self-heals schema drift in CSV files using rules and a local LLM.

- Developed a 4-task event-driven Airflow pipeline to validate, load and transform CSVs, with 2 retries per task.
- Implemented LLaMA 3.2 self-healing that maps renamed columns, with 9 checks on every AI answer before use.
- Logged every repair attempt in PostgreSQL and containerized the stack so it starts with one Docker command.

VOCA – Vocabulary Learning Platform Live |
Tech: LangChain, Next.js 14, MongoDB

Vocabulary learning platform for storing, organizing, and practicing English words with interval-based reviews, progress tracking, and AI-assisted support.

- Created a streaming LangChain chat tutor on NVIDIA NIM with saved multi-thread history, where words the AI mentions can be added to the library in one tap.
- Secured 15 REST API routes by scoping every query to the signed-in user, with input whitelisting, NoSQL-injection guards and rate limiting on login and chat.
- Generated AI example sentences and mnemonics for each word, and auto-filled meanings from a dictionary API.

Technical Skills

Languages & CS: C++, Python, Data Structures & Algorithms
Backend & APIs: FastAPI, REST APIs, Microservices, API Testing, pytest, Redis
Databases: SQL (MySQL, PostgreSQL), MongoDB, ChromaDB
AI & Automation: GenAI, LangChain, LLM Integration, Ollama, Streamlit
Machine Learning: Pandas, NumPy, Scikit-learn, Regression & Classification, Neural Networks, NLP
Cloud & DevOps: Git, GitHub, Docker

Certifications & Achievements

Real-Time Detection of Halo Coronal Mass Ejections from Solar Wind Data 2026
Research paper (co-author), published at UPES ICACSDF 2026

Amazon ML Summer School 2026
Selected nationwide for Amazon's competitive ML training program

Advanced Learning Algorithms Dec 2025
DeepLearning.AI & Stanford University (Coursera)

Supervised Machine Learning: Regression and Classification Jan 2025
DeepLearning.AI & Stanford University (Coursera)